

Dataset	Noise Type	Noisy Thought Num	Accuracy
NoRa-Math Base-9	clean	0 (total thoughts per shot: 8)	46.4
	irrelevant	1 (9)	46.4
		2 (10)	40.0
		3 (11)	34.2
		4 (12)	30.5
		5 (13)	28.8
		6 (14)	25.3
	inaccurate	1	34.8
		2	24.8
		3	15.1
		4	11.1
		5	6.8
		6	6.2
NoRa-Symbolic Equal	clean	0 (total thoughts per shot: 10.2)	39.8
	irrelevant	1 (11.2)	32.8
		2 (12.2)	34.2
		3 (13.2)	33.3
		4 (14.2)	30.6
		5 (15.2)	28.4
		6 (16.2)	29.2
	inaccurate	1	34.5
		2	31.3
		3	32.4
		4	31.4
		5	28.3
		6	27.3

Table 39: Impact of the number of noisy thoughts on GPT-3.5 performance.

The experimental results demonstrate that the presence of noisy thoughts, even in small numbers, can significantly impact the model’s reasoning performance. The extent of the impact depends on factors such as the task type, noise type, and number of noisy thoughts.

F.6 The Number of Noisy Examples

In this part, we provide a deeper analysis and extract more insights into in-context learning with varying numbers of noisy examples.

First, we obtain the accuracy of the base model with different numbers of noisy prompting examples, including 0-shot, 1-shot, 3-shot, and 5-shot. According to the results shown in Tab. 40, we propose the following hypothesis: *The trend of model performance as the number of noisy examples increases depends on the change in performance between 1-shot and 0-shot learning.*

Namely, if the performance in 1-shot learning is better than in 0-shot learning, this indicates that noisy examples still help the model in recognizing task patterns. Therefore, model performance can improve with an increasing number of noisy examples, albeit at a slower rate compared to clean examples. On the other hand, if the performance in one-shot learning is less than in zero-shot learning, it suggests that noisy examples interfere with the model’s existing knowledge of the task, causing model performance to decline as the number of noisy examples increases.

Task	Setting	#Prompting Examples			
		0-shot	1-shot	3-shot	5-shot
Base-9	ina.-easy	7.2	17.5	23.2	25.6
	ina.-hard	7.2	11.3	6.0	5.7
Base-11	irr. easy	5.5	8.9	19.1	26.3
	irr. hard	5.5	7.7	10.7	16.1
Sym.(E)	ina.-easy	8.8	17.3	29.1	—
	ina.-hard	8.8	15.0	22.7	—
Sym.(L)	irr. easy	0.0	2.3	7.0	8.9
	irr. hard	0.0	1.9	6.0	—

Table 40: Performances of base model under varying number of examples ("—" denotes over token limit, bolded number denotes the highest accuracy).

Next, we conduct a further quantitative analysis of the impact of noisy shots on model performance. We evaluate (0-shot, 1-shot, 3-shot) results partition, there are 8 possible results in total: (CCC), (CCW), (CWC), (CWW), (WCC), (WCW), (WWC), (WWW). Specifically, 'C' stands for correct and 'W' stands for wrong. For example, CCC means the cases in which test questions are correctly answered in 0-shot (C), 1-shot (C), and 3-shot (C). WWC means the cases in which test questions are wrongly answered in 0-shot (W), 1-shot (W), and correctly answered in 3-shot (C).

To analyze the effects of noisy shots, we categorize these eight results into three types:

- The model answers correctly at zero-shot but starts to answer incorrectly as noisy shots increase. The corresponding results are (CCW), (CWC), and (CWW).
- The model answers incorrectly at zero-shot but starts to answer correctly as noisy shots increase. The corresponding results are (WCC), (WCW), and (WWC).
- The model’s performance is consistent at zero-shot and with noisy shots. The corresponding results are (CCC) and (WWW).

Task	Setting	#Result Partition							
		CCC	CCW	CWC	CWW	WCC	WCW	WWC	WWW
Base-9	ina.-easy	0.006	0.01	0.019	0.037	0.049	0.11	0.158	0.611
	ina.-hard	0.003	0.01	0.004	0.055	0.012	<u>0.088</u>	<u>0.041</u>	0.786
Base-11	irr. easy	0.004	0.003	0.013	0.036	0.029	0.054	0.144	0.718
	irr. hard	0.001	0.001	0.008	0.043	0.005	0.069	<u>0.075</u>	0.797
Sym.(E)	ina.-easy	0.053	0.008	0.012	0.015	0.067	0.045	0.16	0.641
	ina.-hard	0.049	0.007	0.015	0.016	0.042	0.052	<u>0.121</u>	0.698
Sym.(L)	irr. easy	0.0	0.0	0.0	0.0	0.005	0.018	<u>0.058</u>	0.919
	irr. hard	0.0	0.0	0.0	0.0	0.004	0.015	<u>0.056</u>	0.925

Table 41: Results partition of (0-shot, 1-shot, 3-shot).

By observing Tab. 41, we arrive at the following analysis.

Observation F.10 (Compared to zero-shot, models can benefit from noisy shots in many cases). Across all datasets, (WWW) is the most prevalent outcome, while the second most common results are almost always (WWC) or (WCW). This implies that there is a relatively large proportion of samples for which the model answers incorrectly at zero-shot, yet correctly as noisy shots increase.

Observation F.11 (The instances where noisy shots have a negative effect are very rare.). For the first category of results, namely (CCW), (CWC), and (CWW), the average proportion across all datasets is 0.039. This indicates that the proportion of samples where the model answers correctly at zero-shot but incorrectly as noisy shots increase is very small. In other words, the instances where noisy shots have a negative effect are rare.

Observation F.12 (There are significantly more instances where the model benefits from increasing noisy shots). For the second category of results, which includes (WCC), (WCW), and (WWC), the average proportion across all datasets is 0.186, which is greater than that of the first category. This means that there is a larger proportion of samples where the model answers incorrectly at zero-shot but correctly as noisy shots increase. Namely, there are more instances where noisy shots have a positive effect.

Observation F.13 (The type of instances where the model produces consistent answers is dominant across all datasets). For the third category of results, namely (CCC) and (WWW), the average proportion across all datasets is the highest, at 0.776. This category of results does not allow for an assessment of the effect of noisy shots.

We also provide a case study of in-context learning using a varying number of noisy examples. We provide cases corresponding to some possible outcomes (*i.e.*, [(CCC), (CCW), (CWC)]) for (0-shot, 1-shot, 3-shot) on the NoRa-Math Base-9 dataset with inaccurate easy-level noise in Tables. 42-44.

F.7 Variants of CD-CoT

Acquiring a human-annotated and clean rationale can be challenging in some practical scenarios. However, we would further clarify the relatively low cost and the necessity of this clean rationale to guarantee the effectiveness of robust reasoning.

Current LLMs cannot discern and remove noisy thoughts with their intrinsic denoising ability, even when enhanced with the self-denoising method. This is supported by the empirical observations

Question

In base-9, what is 25+18? Please reason it step by step. End the response with the result in "Answer: result".

0-shot (Correct)

To perform addition in base-9, we need to first understand the place values. In base-9, the place values range from right to left as follows: ones, nines, eighty-ones, and so on. Let's start with the ones place. 5 plus 8 is equal to 13. However, in base-9, we can only have digits from 0 to 8. Therefore, we need to regroup. Since 13 is greater than 9, we can subtract 9 from it and carry over 1 to the next place value (nines place). The remainder is 4. Now let's move on to the nines place. We have 1 (carry over) plus 2 plus 1 (from the nines place) which is equal to 4. Therefore, when we add 25 and 18 in base-9, we get 44. Answer: 44.

1-shot (Correct)

In base-9, the digits are "012345678". We have $5 + 8 = 13$ in base-10. Since we're in base-9, that exceeds the maximum value of 8 for a single digit. $13 \bmod 9 = 4$, so the digit is 4 and the carry is 1. Next, we have $2 + 1 + 1$ (carry) = 4 in base-10. 4 in base-9 is written as 4. Since there's no carry this time, we have no additional digits. So, the sum is 44 in base-9. Answer: 44.

3-shot (Correct)

In base-9, the digits are "012345678". First, we add the digits in the rightmost place value, which is $5 + 8$. We have $5 + 8 = 13$ in base-10. Since we're in base-9, the maximum value for a single digit is 8. $13 \bmod 9 = 4$, so the digit is 4 and the carry is 1. Next, we add the digits in the next place value, which is 1 (from the carry) + $2 + 1$. We have $1 + 2 + 1 = 4$ in base 10. Again, $4 \bmod 9 = 4$, so the digit is 4 and there is no carry. Therefore, the final result is 44. Answer: 44.

Table 42: Case for (CCC): 0-shot correct, 1-shot correct, 3-shot correct.

Question

In base-9, what is 57+26? Please reason it step by step. End the response with the result in "Answer: result".

0-shot (Correct)

To add in base-9, we start by adding the digits in the rightmost column. In the units column, $7 + 6 = 13$. Since we are in base-9, we need to regroup. We can rewrite 13 as 4 in the units column and carry-over 1 to the next column. In the next column, we add the carried-over 1 to the digits in that column. We have $1 + 5 + 2 = 8$. So, the result in the next column is 8. Finally, we put the digits 8 and 4 together to get the final answer. Answer: 84.

1-shot (Correct)

To add 57 and 26 in base-9, we'll start with the units digit. In base-9, the digits are "012345678". $7 + 6 = 13$. Since 13 is greater than 8 (the maximum digit value in base-9), we need to regroup. We can write 13 as $13 = 1 * 9 + 4$. Therefore, the units digit is 4 and there is a carry of 1 to the next column. Moving on to the next column, we have $5 + 2 + 1$ (the carried digit) = 8. 8 is a valid digit in base-9, so we do not need to regroup. The tens digit is 8. Therefore, the sum of 57 and 26 in base-9 is 84. Answer: 84.

3-shot (Wrong)

In base-9, the digits are "012345678". We have $7 + 6 = 13$. Since we're in base-9, we need to determine the digit and carry. 13 divided by 9 is 1 with a remainder of 4. So the digit is 4, and we carry 1. Next, we have $5 + 2 + 1 = 8$. So the digit is 8 and there is no carry. A leading digit is not needed since there is no carry. Thus, the answer is 48. Answer: 48.

Table 43: Case for (CCW): 0-shot correct, 1-shot correct, 3-shot wrong.

in Sec. 4. Hence, external supervision is necessary, which should be sufficient for denoising and accessible in practice. Existing methods with external supervision require (1) oracle feedback on the test question, (2) human feedback of errors on specific tokens or positions, or (3) expert knowledge to construct detailed descriptions of specific tasks.

By contrast, we believe that a clean CoT demonstration is more attainable and practical, which can be the minimal requirement for denoising-purpose prompting. With the proposed CD-CoT method, we show that LLMs can identify noisy thoughts by contrasting a noisy rationale with a clean one.

Besides, we build a self-supervised variant of CD-CoT that does not rely on manually crafted clean examples, *i.e.*, this variant can get a clean rationale automatically. Specifically, we prompt LLM with a fixed question using the zero-shot CoT approach [39], allowing it to answer N times. We then select the responses with the correct answer based on the ground-truth label.

Then, we conduct experiments on two datasets, NoRa-Math Base-9 and NoRa-Symbolic Equal, with the default model (GPT-3.5-0613). The hyper-parameters are temperature=0.1 and top-p=1. We set $N=50$ for NoRa-Math Base-9 and $N=100$ for NoRa-Symbolic Equal. Other settings are kept the same as default with the original CD-CoT.